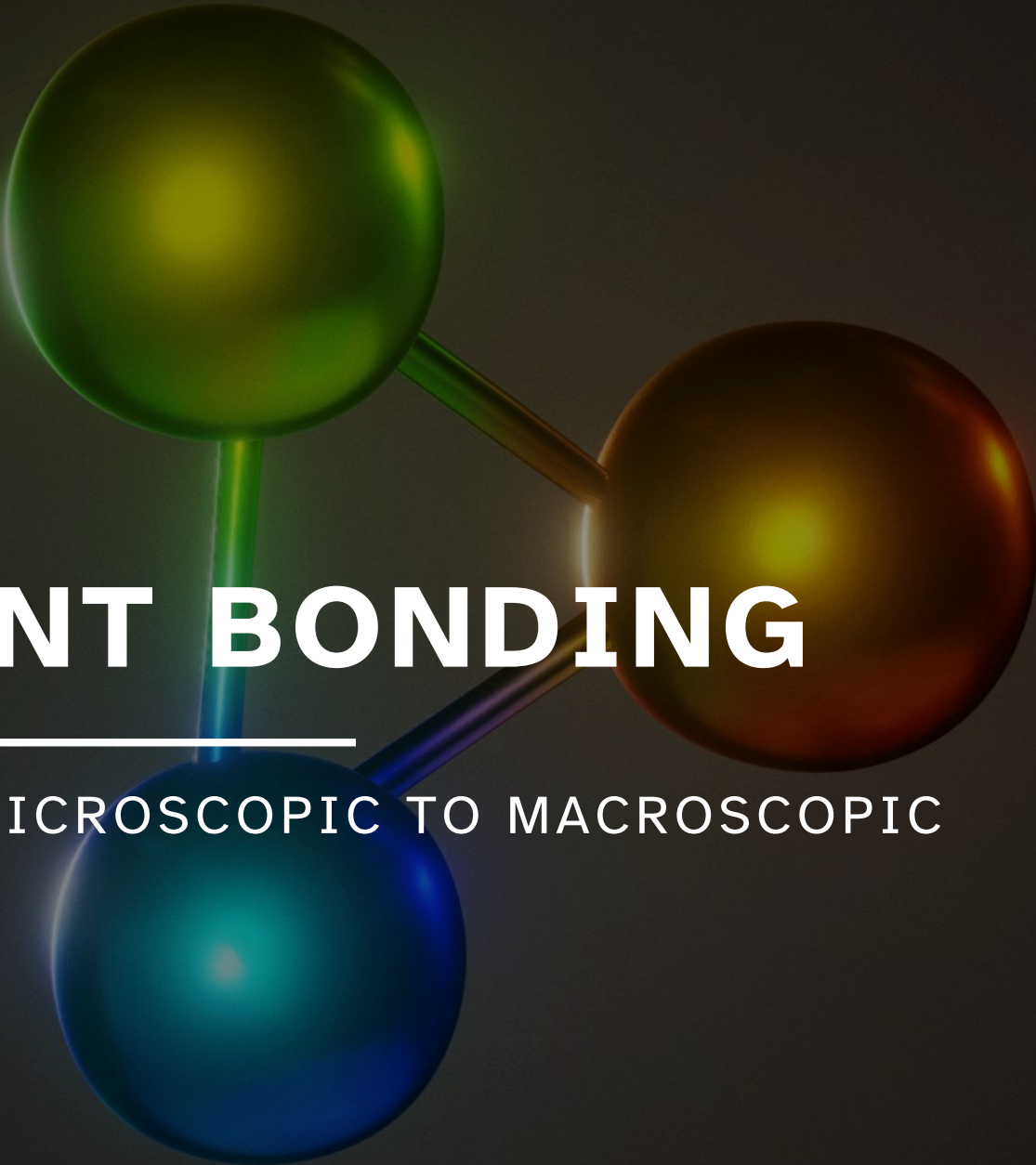




10A COVALENT BONDING

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 10

1. Predict the number of bonds an atom is likely to form in a molecule.
2. Draw Lewis structures for molecules that follow the octet rule.
3. Define and apply electronegativity and its periodic trends.
4. Draw Lewis structures for molecules for which multiple valid Lewis structures exist (called resonance).
5. Draw Lewis structures for molecules that do not obey the octet rule by application of the typical exceptions exhibited by H, Be, B, and expanded octets.
6. Determine bond polarity using electronegativity values.
7. Describe and apply the relation between bond length and bond strength.
8. Apply bond enthalpies to calculate enthalpies of reaction.

Where did we start and where are we going?

Chemistry in Context

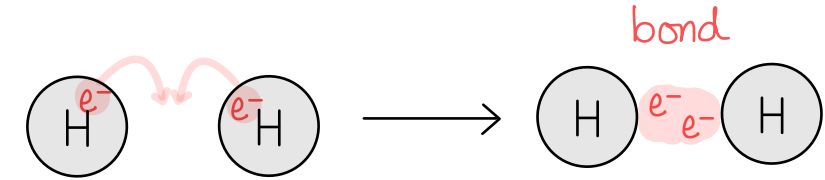
What is our overall goal?



Formation of Covalent Bonds

10.1

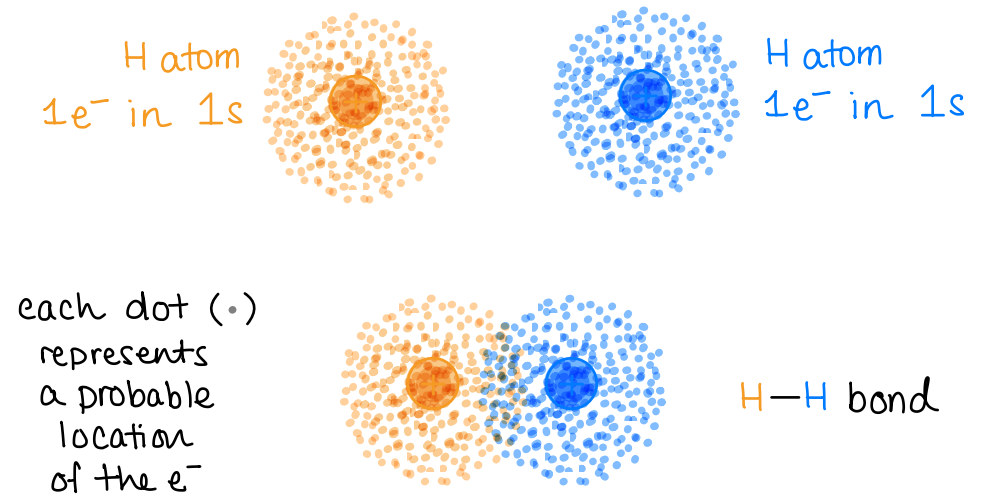
Covalent bonds



- Formed by the *sharing of electrons* in the space between two atoms.

How is it possible for these electrons to count for both atoms involved in the covalent bond?

- Electrons are small (quantum) particles.
- Electron properties have *wave-like* probabilities.
- Electron position can be thought of as “smeared” 3D charge clouds called orbitals.
- Electrons are *shared* when the orbitals (or “smeared” electron charge clouds) *overlap*.



Formation of Covalent Bonds

10.1

Octet (8 e⁻) rule

- Atoms with **eight valence electrons are particularly stable** because they have a **full valence shell** with an electron configuration that is: $s^2 p^6$
- In a molecule, atoms will form covalent bonds so that each atom attains a *complement* of 8 electrons THROUGH SHARING.

especially true for n=2 row, but more a suggestion than a rule

Duet (2 e⁻) rule

- For H and He, the valence shell is $n = 1$ and can only hold two electrons.
- They can share *only* two electrons in molecules to achieve a $1s^2$ configuration.

Resonance and Electronegativity

10.3 (we are not covering formal charges)

Electronegativity (EN)

- The ability of an atom to attract *shared* electrons to itself in a bond.
- Different numerical scales exist, but Pauling scale is most common.
- Trends in EN values depend on many factors: size, e^- config, IE , Z_{eff} , etc.

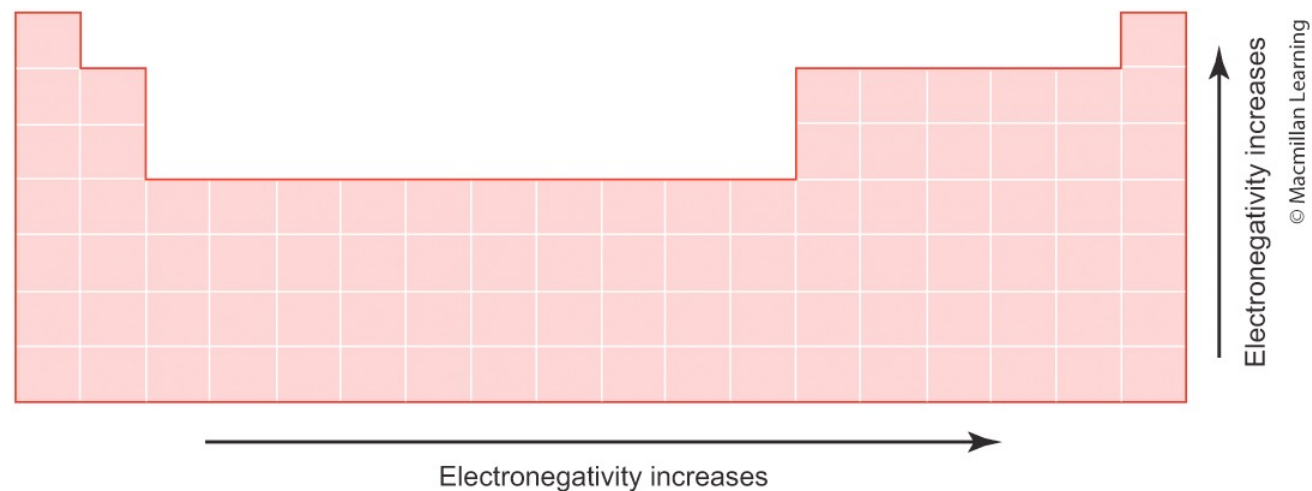
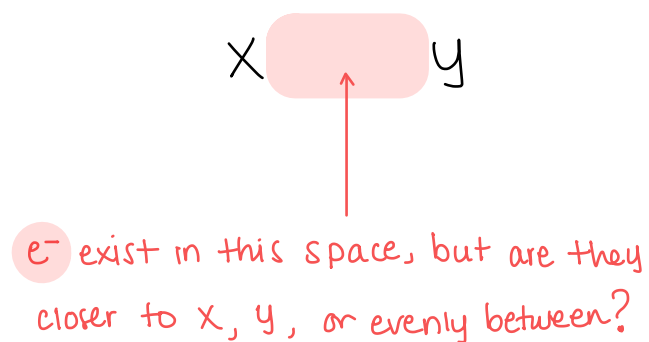


FIGURE 11.15 General electronegativity trend

Resonance and Electronegativity

10.3 (we are not covering formal charges)

An element with a *high electronegativity*:

- Pulls or attracts shared electrons closer to itself than an element with a lower electronegativity in a bond.
- less likely to share e^-
- less likely to form many bonds

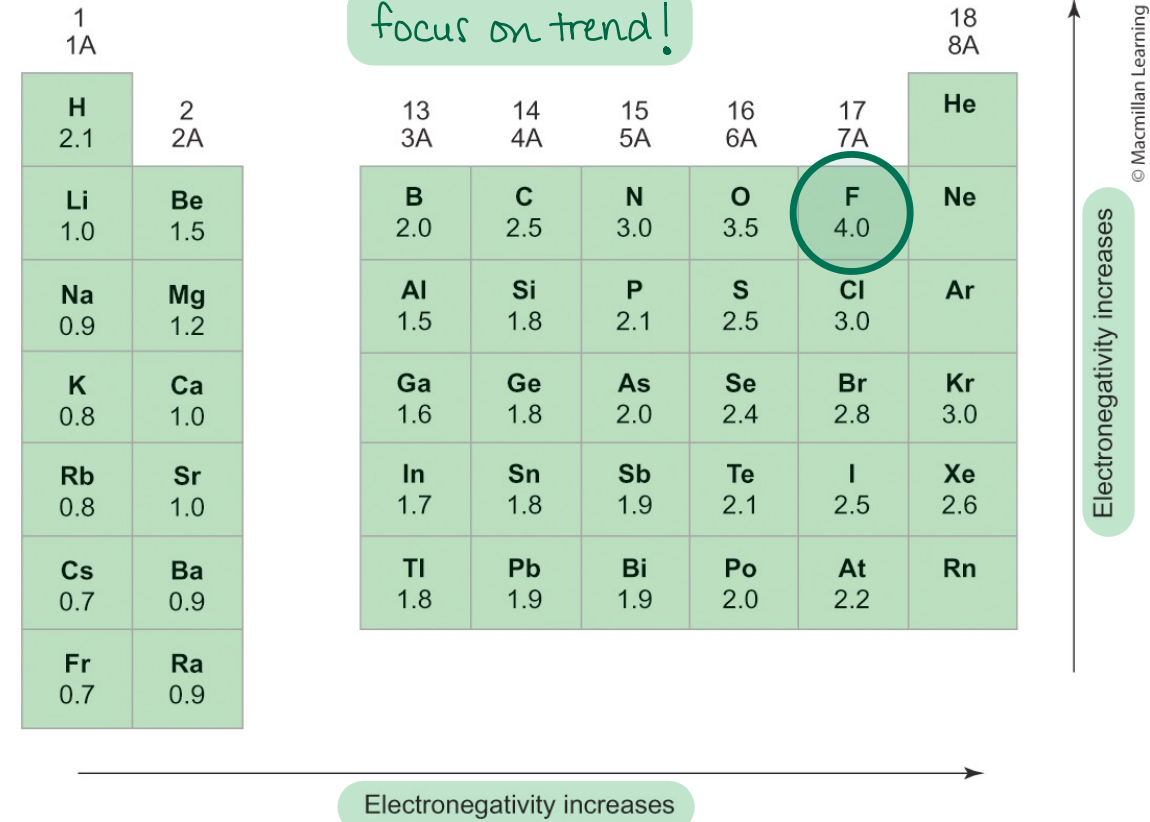


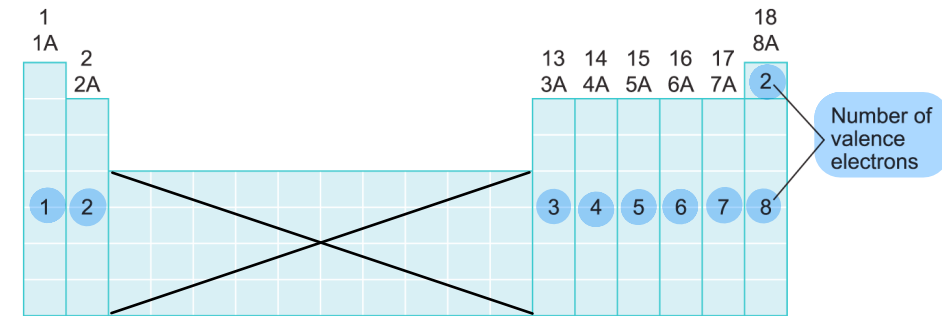
FIGURE 10.7 Pauling scale for electronegative values

Lewis Structures

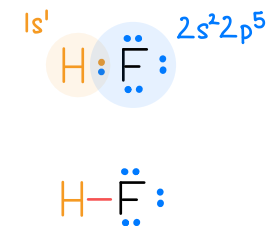
10.2

Lewis structure: *localized electron* model for the structure of covalent molecules

- **Only valence electrons**
- Total number of e^- in Lewis structure *must equal* total number of valence e^- from all atoms
- Satisfy octet ($8 e^-$) and duet ($2 e^-$) rules



Atom	# of valence e^-	# of shared bonding e^-	# of unshared lone pair e^-	Total # of e^- around atom
Hydrogen H	1	2	0	2
Fluorine F	7	2	6	8



Formation of Covalent Bonds

10.1

Atoms can *share* different numbers of electrons to form different types of bonds.

Number of <i>shared</i> e^-	Bond Type	Representation	Example
2	Single bond	—	H_2, F_2
4	Double bond	=	O_2
6	Triple bond	≡	N_2

notes: area between atoms is shared;
each line is $2e^-$;
atoms end up with ns^2np^6 configuration



14 valence e^-



12 valence e^-



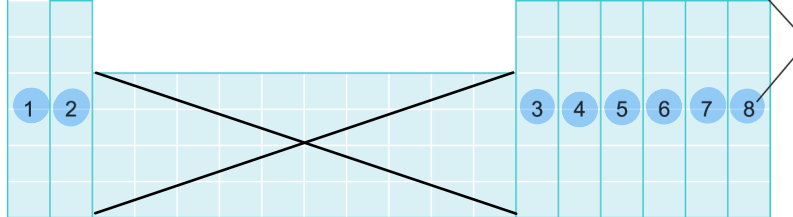
10 valence e^-

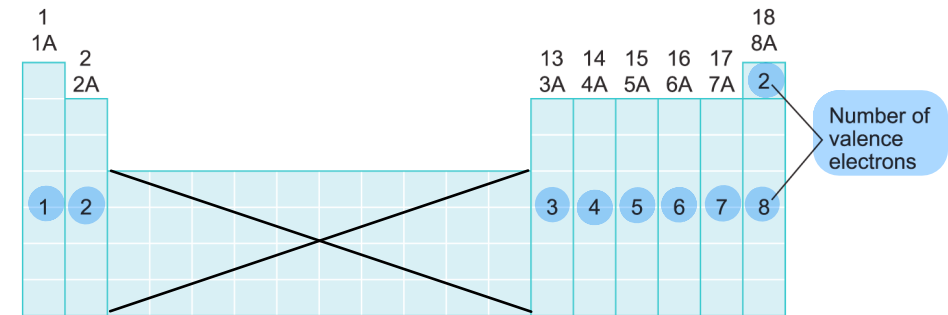
Lone pair: Set of $2e^-$ that are *not shared* (not involved in bonding).

Lewis Structures

10.2

Method 1: Electron-Pushing

1. Count total number of valence e^- from all atoms.
 2. Assign central atom and arrange all other atoms around it.
 - Central atom is *never* hydrogen.
 - Often: central atom is written *first* in the chemical formula, or it's the *least electronegative* atom.
 3. Make skeleton: Use single bonds (—) to connect central atom to outer atoms.
 4. Add lone pairs ($\cdot\cdot$) to outer atoms to satisfy octet rule, or duet rule for hydrogen.
 5. Satisfy octet rule for central atom.
 - a) If leftover e^- (from step 1), add as lone pairs ($\cdot\cdot$) to central atom.
 - b) If *no* leftover e^- (from step 1), re-assign lone pairs from outer atoms to convert single bonds (—) to double bonds (=) or triple bonds ($\equiv$).
 6. Verify Lewis structure is valid: count total number of e^- in structure and check octet/duet rule.
- 



Lewis Structures

10.2

Method 2: Guess-and-Check

1. Count total number of valence e^- from all atoms.
2. Assign central atom and arrange all other atoms around it.
 - Central atom is *never* hydrogen.
 - Often: central atom is written *first* in the chemical formula, or it's the *least electronegative* atom.
3. Make skeleton: Use single bonds (—) to connect central atom to outer atoms.
4. Add lone pairs (·) to outer atoms to satisfy octet rule, or duet rule for hydrogen.
5. Satisfy octet rule for central atom.
6. Count total number of e^- in structure drawn.
 - a) If correct number of electrons as step 1, then valid Lewis structure.
 - b) If *too many electrons* in structure, go back to step 3 and try double (=) and/or triple ($\equiv$) bonds.

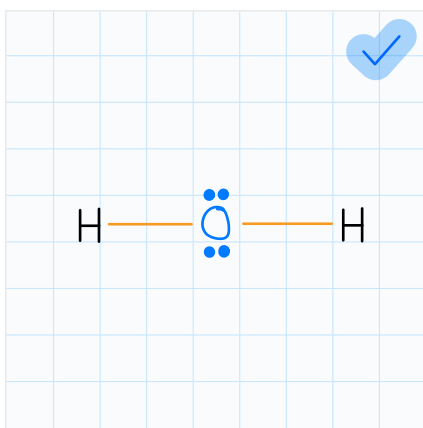
1 1A	2 2A		13 3A	14 4A	15 5A	16 6A	17 7A	18 8A
1	2		3	4	5	6	7	2

Number of valence electrons

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw the Lewis structure for H₂O.



$$\begin{array}{rcl}
 2(1e^-) + 6e^- & = & 8e^- \quad \text{total } e^- \\
 - 4e^- & & 2 \text{ single bonds} \\
 \hline
 4e^- \\
 - 0e^- & & \text{octets on outer atoms} \\
 \hline
 4e^- \\
 - 4e^- & & \text{octet on central atom} \\
 \hline
 0e^-
 \end{array}$$

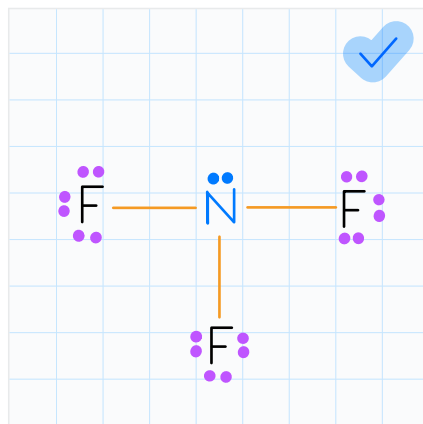
1. Count total # valence e^-
2. Pick central atom
3. Connect to central atoms using single bonds
4. Satisfy octet for outer atoms, or duet for H
5. Satisfy octet for central atom
 - a) If leftover e^- , assign as lone pairs
 - b) If no e^- left over, move lone pairs from outer atoms to double/triple bonds
6. Verify octet/duet rule and correct # total e^-

1 H Hydrogen 1.008																	2 He Helium 4.003		
3 Li Lithium 6.941	4 Be Beryllium 9.012													5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 18.998	10 Ne Neon 20.18
11 Na Sodium 22.99	12 Mg Magnesium 24.31													13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.88	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.38	31 Ga Gallium 69.72	32 Ge Germanium 72.59	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80		
37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium (98)	44 Ru Ruthenium 101.1	45 Rh Rhodium 102.9	46 Pd Palladium 106.4	47 Ag Silver 107.9	48 Cd Cadmium 112.4	49 In Indium 114.8	50 Sn Tin 118.7	51 Sb Antimony 121.8	52 Te Tellurium 127.6	53 I Iodine 126.9	54 Xe Xenon 131.3		
55 Cs Cesium 132.9	56 Ba Barium 137.3	57 La Lanthanum 138.9	72 Hf Hafnium 178.5	73 Ta Tantalum 180.9	74 W Tungsten 183.9	75 Re Rhenium 186.2	76 Os Osmium 190.2	77 Ir Iridium 192.2	78 Pt Platinum 195.1	79 Au Gold 197.0	80 Hg Mercury 200.6	81 Tl Thallium 204.4	82 Pb Lead 207.2	83 Bi Bismuth 209.0	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)		
87 Fr Francium (223)	88 Ra Radium (226)	89 Ac Actinium (227)	104 Rf Rutherfordium (261)	105 Db Dubnium (264)	106 Sg Seaborgium (266)	107 Bh Bohrium (269)	108 Hs Hassium (277)	109 Mt Meitnerium (276)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (280)	112 Cn Copernicium (285)	113 Nh Nihonium (284)	114 Fl Flerovium (289)	115 Mc Moscovium (288)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)		

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw the Lewis structure for NF_3 .



$$5e^- + 3(7e^-) = 26e^- \quad \text{total } e^-$$

$$- 6e^- \quad 3 \text{ single bonds}$$

$$20e^-$$

$$- 18e^- \quad \text{octets on outer atoms}$$

$$2e^-$$

$$- 2e^- \quad \text{octet on central atom}$$

$$0e^-$$

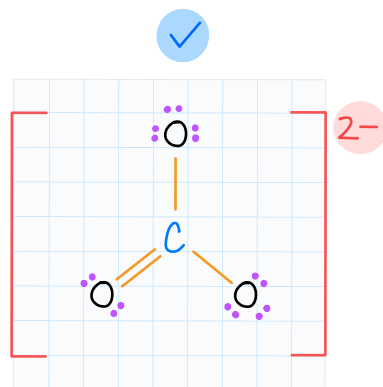
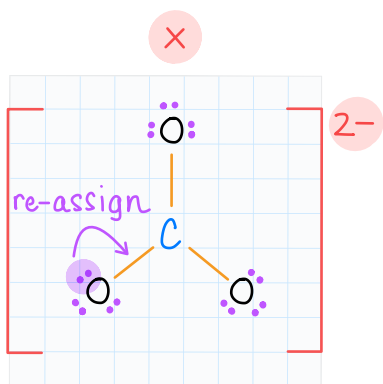
- Count total # valence e^-
- Pick central atom
- Connect to central atoms using single bonds
- Satisfy octet for outer atoms, or duet for H
- Satisfy octet for central atom
 - If leftover e^- , assign as lone pairs
 - If no e^- left over, move lone pairs from outer atoms to double/triple bonds
- Verify octet/duet rule and correct # total e^-

1 H Hydrogen 1.008																	2 He Helium 4.003		
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55 Cs Cesium 132.9	56 Ba Barium 137.3	57 La Lanthanum 138.9	72 Hf Hafnium 178.5	73 Ta Tantalum 180.9	74 W Tungsten 183.8	75 Re Rhenium 186.2	76 Os Osmium 190.2	77 Ir Iridium 192.2	78 Pt Platinum 195.1	79 Au Gold 197.0	80 Hg Mercury 200.6	81 Tl Thallium 204.4	82 Pb Lead 207.2	83 Bi Bismuth 209.0	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)		
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EXAMPLE PROBLEM

Learning Objective(s): 10.2, 10.4

Draw a valid Lewis structure for the carbonate anion (CO_3^{2-}).



for ions:
use square brackets
and put overall charge
or structure assumed
to be neutral!

$$4 + 3(6) + 2 = 24 e^-$$

total e^-

$$- 6 e^-$$

3 single bonds

$$18 e^-$$

$$- 18 e^-$$

$$0 e^-$$

octets on outer atoms

out of e^- to fulfill
octet on central atom!

→ enact step 5b

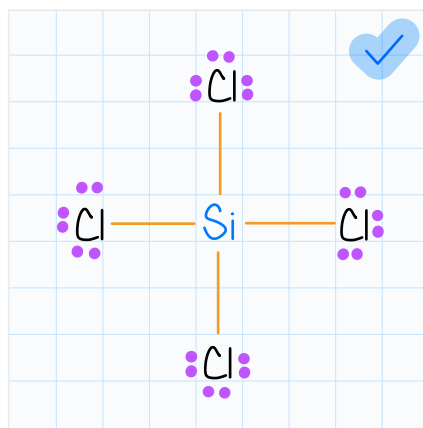
- Count total # valence e^-
- Pick central atom
- Connect to central atoms using single bonds
- Satisfy octet for outer atoms, or duet for H
- Satisfy octet for central atom
 - If leftover e^- , assign as lone pairs
 - If no e^- left over, move lone pairs from outer atoms to double/triple bonds
- Verify octet/duet rule and correct # total e^-

1 H Hydrogen 1.008																	2 He Helium 4.003		
3 Li Lithium 6.941	4 Be Beryllium 9.012													5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18
11 Na Sodium 22.99	12 Mg Magnesium 24.31													13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.88	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.38								
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IN-CLASS QUESTION 1

Learning Objective(s): 10.2, 10.3

Draw the Lewis structure for silicon tetrachloride (SiCl_4).



$$\begin{array}{rcl} 4e^- + 4(7e^-) & = & 32e^- & \text{total } e^- \\ - 8e^- & & & \text{4 single bonds} \\ \hline 24e^- & & & \\ - 24e^- & & & \text{octets on outer atoms} \\ \hline 0e^- & & & \end{array}$$

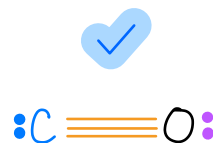
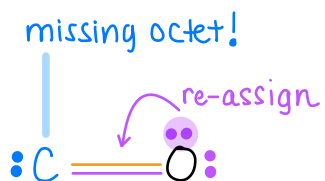
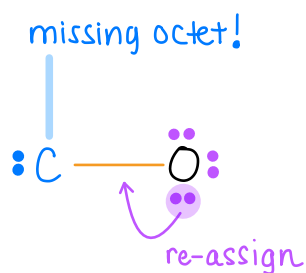
What is the total number of valence electrons in your Lewis structure? $32 e^-$

What is the identity of the central atom? Si

IN-CLASS QUESTION 2

Learning Objective(s): 10.1, 10.2

Draw the Lewis structure for carbon monoxide (CO).



$$\begin{array}{rcl}
 4e^- + 6e^- & = & 10e^- \quad \text{total } e^- \\
 - 2e^- & & | \text{ single bond} \\
 \hline
 8e^- \\
 - 6e^- & & \text{octets on outer atom} \\
 \hline
 2e^- & & \text{out of } e^- \text{ to fulfill} \\
 & & \text{octet on central atom!} \\
 & & \rightarrow \text{enact step 5b twice}
 \end{array}$$

Complete the table below to describe the structure and bonding.

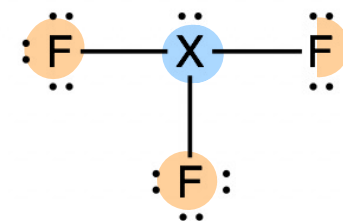
Atom	# of valence e^-	# of shared bonding e^-	# of unshared lone pair e^-	Total # of e^- around atom
Carbon C	4	6	2	8
Oxygen O	6	6	2	8

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Which elements could be X in the Lewis structure shown?

Select all that apply.



☐ C

☒ N

☐ O

☐ F

☐ Si

☒ P

$$(\text{valence } e^- \text{ on } X) + 3(7 e^-) = 26 \text{ total } e^-$$

$$\text{valence } e^- \text{ on } X = 5 e^-$$

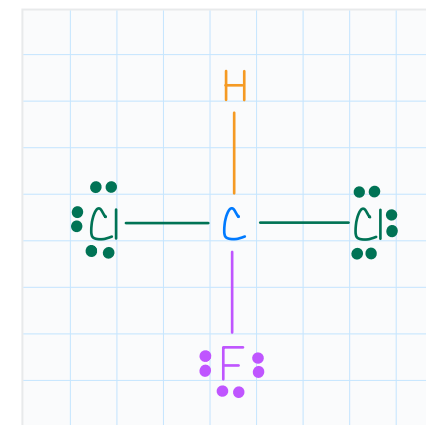
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EXAMPLE PROBLEM

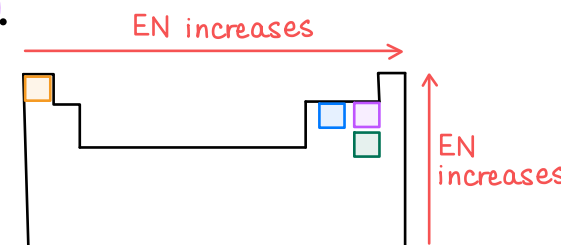
Learning Objective(s): 10.2, 10.3

Draw the Lewis structure for CHCl_2F . $4e^- + 1e^- + 2(7e^-) + 7e^- = 26 \text{ total } e^-$

Which statement is true regarding the atoms in this molecule?



- ☐ In the bond between carbon and hydrogen, hydrogen attracts electrons more strongly than carbon.
- ☐ In the bond between carbon and chlorine, carbon gets a larger share of the electron pair than chlorine.
- ☒ Fluorine has the ^{largest EN} greatest ability to attract electrons to itself in a bond.
- ☐ Carbon is the most electronegative atom.

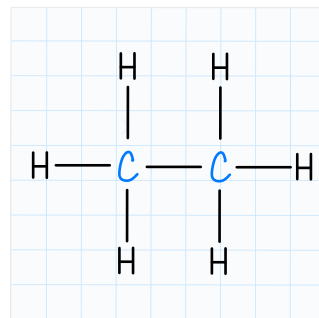


EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw the Lewis structure for C_2H_6 .

$$2(4e^-) + 6(1e^-) = 14e^-$$



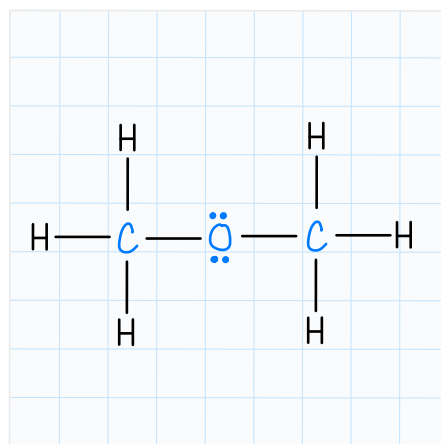
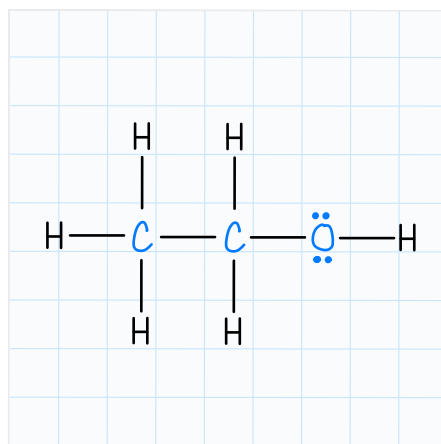
this is only possible structure!
we call both C atoms "central" atoms!

1 H Hydrogen 1.008																	2 He Helium 4.003														
3 Li Lithium 6.941	4 Be Beryllium 9.012													5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18												
11 Na Sodium 22.99	12 Mg Magnesium 24.31													13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95												
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.88	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.38	31 Ga Gallium 69.72	32 Ge Germanium 72.59	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80														
37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium (99)	44 Ru Ruthenium 101.1	45 Rh Rhodium 102.9	46 Pd Palladium 106.4	47 Ag Silver 107.9	48 Cd Cadmium 112.4	49 In Indium 114.8	50 Sn Tin 118.7	51 Sb Antimony 121.8	52 Te Tellurium 127.6	53 I Iodine 126.9	54 Xe Xenon 131.3														
55 Cs Cesium 132.90	56 Ba Barium 137.3	57 La Lanthanum 138.9	58 Ce Cerium 140.1	59 Pr Praseodymium 140.9	60 Nd Neodymium 144.2	61 Pm Promethium (145)	62 Sm Samarium 150.4	63 Eu Europium 151.9	64 Gd Gadolinium 157.3	65 Tb Terbium 158.9	66 Dy Dysprosium 162.5	67 Ho Holmium 164.9	68 Er Erbium 167.3	69 Tm Thulium 168.9	70 Yb Ytterbium 173.0	71 Lu Lutetium 174.9	72 Hf Hafnium 178.5	73 Ta Tantalum 180.9	74 W Tungsten 183.8	75 Re Rhenium 186.2	76 Os Osmium 190.2	77 Ir Iridium 192.2	78 Pt Platinum 195.1	79 Au Gold 197.0	80 Hg Mercury 200.6	81 Tl Thallium 204.4	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
87 Fr Francium (223)	88 Ra Radium (226)	89 Ac Actinium (227)	90 Th Thorium (232)	91 Pa Protactinium (231)	92 U Uranium (238)	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (262)	104 Rf Rutherfordium (261)	105 Db Dubnium (262)	106 Sg Seaborgium (266)	107 Bh Bohrium (264)	108 Hs Hassium (277)	109 Mt Meitnerium (268)	110 Ds Darmstadtium (271)	111 Rg Roentgenium (272)	112 Cn Copernicium (285)	113 Nh Nihonium (284)	114 Fl Flerovium (289)	115 Mc Moscovium (288)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw two *different* Lewis structures for a molecule with the formula $\text{C}_2\text{H}_6\text{O}$.



$$2(4e^-) + 6(1e^-) + 6e^- = 20e^-$$

These are two different compounds with different properties!

Periodic table showing elements and their atomic numbers.

BONUS QUESTION

Assessed on: Summer 2021, Exam 2

Draw the Lewis structures for CO_2 , H_2O , and O_2 .

- a) Which molecule has the greatest number of electrons in shared bonds?

- b) Which molecule has the smallest number of lone pair electrons?